KAIZEN MATHS

Lesson starter

Basic Integration and Definite Integrals

Trapezium Rule

Name _____

S1

Use the trapezium rule with $n = 4$ strips to approximate $\int_0^5 (3x^2 - 2x + 0)\,dx$.

S2

Use the table and the trapezium rule to approximate $\int_1^5 y\,dx$ with $n = 4$ strips.

S3

Use the trapezium rule with $n = 4$ strips to approximate $\int_0^3 (2x^2 - 3x + 2)\,dx$.

S4

Use the table and the trapezium rule to approximate $\int_2^5 y\,dx$ with $n = 6$ strips.

Show your working

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S4

Use the table and the trapezium rule to approximate $\int_2^5 y\,dx$ with $n = 6$ strips.

ALG-06EF-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Trapezium Rule

Name _____

E1

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^3 (3x^2 + 3x + 0)\,dx$.

E2

Use the table and the trapezium rule to approximate $\int_1^3 y\,dx$ with $n = 4$ strips.

E3

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^4 (1x^2 - 1x - 1)\,dx$.

Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Trapezium Rule

Name _____

E1

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^3 (3x^2 + 3x + 0)\,dx$.

E2

Use the table and the trapezium rule to approximate $\int_1^3 y\,dx$ with $n = 4$ strips.

E3

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^4 (1x^2 - 1x - 1)\,dx$.

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Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Trapezium Rule

Name _____

E1

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^3 (3x^2 + 3x + 0)\,dx$.

E2

Use the table and the trapezium rule to approximate $\int_1^3 y\,dx$ with $n = 4$ strips.

E3

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^4 (1x^2 - 1x - 1)\,dx$.

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Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Trapezium Rule

Name _____

E1

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^3 (3x^2 + 3x + 0)\,dx$.

E2

Use the table and the trapezium rule to approximate $\int_1^3 y\,dx$ with $n = 4$ strips.

E3

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^4 (1x^2 - 1x - 1)\,dx$.

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Show your working

Teacher answers & working

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S1

Use the trapezium rule with $n = 4$ strips to approximate $\int_0^5 (3x^2 - 2x + 0)\,dx$.

Answer:

≈ 103.91

Working:

Step 1: Split the interval into equal widths and calculate the y-values from the function.

Step size: $h = (5 - 0)/4 = 1.250$.

i	x_i	y_i
0	0.000	0.000
1	1.250	2.188
2	2.500	13.750
3	3.750	34.688
4	5.000	65.000

$T = \frac{h}{2}(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 103.91$.

Final answer: ≈ 103.91

Teacher answers & working

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S2

Use the table and the trapezium rule to approximate $\int_{1}^{5} y\,dx$ with $n = 4$ strips.

Answer:

≈ 62.00

Working:

Step 1: Read the equal spacing and y-values from the table, then substitute into the trapezium formula.

Confirm $h = (5-1)/4 = 1.000$.

i	x_i	y_i
0	1.000	4.000
1	2.000	8.000
2	3.000	14.000
3	4.000	22.000
4	5.000	32.000

Use $T = \frac{h}{2}(y_0 + 2\sum y_i + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 62.00$.

Final answer: ≈ 62.00

Teacher answers & working

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S3

Use the trapezium rule with $n = 4$ strips to approximate $\int_0^3 (2x^2 - 3x + 2)\,dx$.

Answer:

≈ 11.06

Working:

Step 1: Split the interval into equal widths and calculate the y-values from the function.

Step size: $h = (3 - 0)/4 = 0.750$.

i	x_i	y_i
0	0.000	2.000
1	0.750	0.875
2	1.500	2.000
3	2.250	5.375
4	3.000	11.000

$T = \frac{h}{2}(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 11.06$.

Final answer: ≈ 11.06

Teacher answers & working

Basic Integration and Definite Integrals · Trapezium Rule

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S4

Use the table and the trapezium rule to approximate $\int_{2}^{5} y\,dx$ with $n = 6$ strips.

Answer:

≈ 63.13

Working:

Step 1: Read the equal spacing and y-values from the table, then substitute into the trapezium formula.

Confirm $h = (5-2)/6 = 0.500$.

i	x_i	y_i
0	2.000	9.000
1	2.500	12.250
2	3.000	16.000
3	3.500	20.250
4	4.000	25.000
5	4.500	30.250
6	5.000	36.000

Use $T = \frac{h}{2}(y_0 + 2\sum y_i + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 63.13$.

Final answer: ≈ 63.13

Teacher answers & working

Basic Integration and Definite Integrals · Trapezium Rule

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E1

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^3 (3x^2 + 3x + 0)\,dx$.

Answer:

≈ 40.88

Working:

Step 1: Split the interval into equal widths and calculate the y-values from the function.

Step size: $h = (3 - 0)/6 = 0.500$.

i	x_i	y_i
0	0.000	0.000
1	0.500	2.250
2	1.000	6.000
3	1.500	11.250
4	2.000	18.000
5	2.500	26.250
6	3.000	36.000

$T = \frac{h}{2}(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 40.88$.

Final answer: ≈ 40.88

Teacher answers & working

Basic Integration and Definite Integrals · Trapezium Rule

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E2

Use the table and the trapezium rule to approximate $\int_{1}^{3} y\,dx$ with $n = 4$ strips.

Answer:

≈ 2.75

Working:

Step 1: Read the equal spacing and y-values from the table, then substitute into the trapezium formula.

Confirm $h = (3-1)/4 = 0.500$.

i	x_i	y_i
0	1.000	0.000
1	1.500	0.250
2	2.000	1.000
3	2.500	2.250
4	3.000	4.000

Use $T = \frac{h}{2}(y_0 + 2\sum y_i + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 2.75$.

Final answer: ≈ 2.75

Teacher answers & working

Basic Integration and Definite Integrals · Trapezium Rule

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E3

Use the trapezium rule with $n = 6$ strips to approximate $\int_0^4 (1x^2 - 1x - 1)\,dx$.

Answer:

≈ 9.63

Working:

Step 1: Split the interval into equal widths and calculate the y-values from the function.

Step size: $h = (4 - 0)/6 = 0.667$.

i	x_i	y_i
0	0.000	-1.000
1	0.667	-1.222
2	1.333	-0.556
3	2.000	1.000
4	2.667	3.444
5	3.333	6.778
6	4.000	11.000

$T = \frac{h}{2}(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n)$.

Add the interior ordinates.

Substitute into the trapezium rule.

Calculate the bracket first.

$T \approx 9.63$.

Final answer: ≈ 9.63